Midterm II PRACTICE Exam Solution

MA 341
03/10/2023 **Name:** ____________________
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Read all of the following information before starting the exam:

- Complete each problem. Show all of your work clearly and in order and justify your answers, as partial credit will be given when appropriate and there may be NO credit given for problems without supporting work.
- **Circle or otherwise indicate your final answers.** All answers should be completely simplified, unless otherwise stated.
- **You may use a *non graphical* calculator.**

1. (*30 points*) Rewrite $y''' + 3y'' + 2y' + y = t$ as system of first order differential equations.

Solution: Rewrite as $y''' = -3y'' - 2y' - y + t$.

$$\text{Denote} \quad x(t) = \begin{pmatrix} y \\ y' \\ y'' \end{pmatrix}, \quad \text{then} \quad x'(t) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{pmatrix} x(t) + \begin{pmatrix} 0 \\ 0 \\ t \end{pmatrix}.$$

2. (*40 points*) Solve

$$x'(t) = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix} x(t) + \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^t$$

subject to IC $x(0) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Solution: The characteristic equation

$$\det(A - \lambda I) = \begin{pmatrix} 2-\lambda & -1 \\ 3 & -2-\lambda \end{pmatrix} = (2-\lambda)(-2-\lambda) + 3 = -(4-\lambda^2) + 3 = \lambda^2 - 1 = 0$$

Thus $\lambda_1 = -1$, and $\lambda_2 = 1$ The associated eigenvectors are solutions to

$$\det(A - \lambda_1 I)v_1 = \begin{pmatrix} 3 & -1 \\ 3 & 1 \end{pmatrix} v_1 = 0 \quad \text{and} \quad \det(A - \lambda_2 I)v_2 = \begin{pmatrix} 1 & -1 \\ 3 & -3 \end{pmatrix} v_1 = 0,$$

i.e. $v_1 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ and $v_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, respectively. Hence, the homogeneous solution is

$$x_h(t) = c_1 \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t.$$

To find x_p we define $\Psi = \begin{pmatrix} e^{-t} & e^t \\ 3e^{-t} & e^t \end{pmatrix}$, which inverse is $\Psi^{-1} = -\frac{1}{2} \begin{pmatrix} e^t & -e^t \\ -3e^{-t} & e^{-t} \end{pmatrix}$

$$x_p(t) = \Psi \int \Psi^{-1} g dt = \Psi \int \begin{pmatrix} -e^{2t} \\ 2 \end{pmatrix} dt = \Psi \begin{pmatrix} -\frac{e^{2t}}{2} \\ 2t \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 4t-1 \\ 4t-3 \end{pmatrix} e^t$$

Thus,

$$x(t) = x_h(t) + x_p(t) = c_1 \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t + \frac{1}{2} \begin{pmatrix} 4t-1 \\ 4t-3 \end{pmatrix} e^t$$

Using ICs

$$x(0) = c_1 \begin{pmatrix} 1 \\ 3 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} -1 \\ -3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

determine that $c_1 = 1/2, and c_2 = 1$.
Finally,

$$x(t) = \frac{1}{2} \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{-t} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t + \frac{1}{2} \begin{pmatrix} 4t-1 \\ 4t-3 \end{pmatrix} e^t$$

3. (*40 points*)
Let

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$

Find the following:

a. A^{-1}

Solution:

$$\left(\begin{array}{ccc|ccc} 2 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_1 \leftarrow R_1 - R_3]{R_2 \leftarrow R_2 - R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & -1 \\ 0 & 1 & -1 & 0 & 1 & -1 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_3 \leftarrow \frac{1}{4}(R_3 - R_1 - R_2)}$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & -1 \\ 0 & 1 & -1 & 0 & 1 & -1 \\ 0 & 0 & 1 & -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{array}\right) \xrightarrow[R_1 \leftarrow R_1 + R_3]{R_2 \leftarrow R_2 + R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{3}{4} & -\frac{1}{4} & -\frac{1}{4} \\ 0 & 1 & 0 & -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} \\ 0 & 0 & 1 & -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{array}\right)$$

Thus, $A^{-1} = \begin{pmatrix} \frac{3}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{pmatrix}$.

b. $|A|$

Solution:

$$\begin{aligned} &= 2\begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} - \begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} \\ &= 2(4-1) - (2-1) + (1-2) = 4 \end{aligned}$$

c. Eigenvalues and eigenvectors of A

Solution:

$$0 = |A - \lambda I| = \begin{vmatrix} 2-\lambda & 1 & 1 \\ 1 & 2-\lambda & 1 \\ 1 & 1 & 2-\lambda \end{vmatrix}$$

$$\begin{aligned} &= (2-\lambda)\begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} - \begin{vmatrix} 1 & 1 \\ 1 & 2-\lambda \end{vmatrix} + \begin{vmatrix} 1 & 2-\lambda \\ 1 & 1 \end{vmatrix} \\ &= (2-\lambda)((2-\lambda)^2 - 1) - ((2-\lambda) - 1) + (1 - (2-\lambda)) \\ &= (2-\lambda)^3 - (2-\lambda) + 2(1 - (2-\lambda)) \\ &= (2-\lambda)^3 - 3(2-\lambda) + 2 \\ &\underset{\mu = 2-\lambda}{=} \mu^3 - 3\mu + 2 \end{aligned}$$

Thus, we need to solve

$$\mu^3 - 3\mu + 2 = 0$$

By trial and error one can easily find that $\mu_1 = -2$ and $\mu_2 = 1$ solves the equation. The third solution can be found using Vietta's theorem that says $\mu_1\mu_2\mu_3 = -2$, which means $\mu_3 = \mu_2 = 1$ is a double root. Finally, $2 - \lambda = -2$ gives $\lambda_1 = 4$, and $2 - \lambda = 1$ gives $\lambda_{2,3} = 1$

To find eigenvectors v_i we need to solve $(A - \lambda_i I)v = 0$, which gives.

$\lambda_1 = 4$

$$(A - 4I)v = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} v = 0$$

$$\begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} \xrightarrow[R_3 \leftarrow 2R_3 + R_1]{R_2 \leftarrow 2R_2 + R_1} \begin{pmatrix} -2 & 1 & 1 \\ 0 & -3 & 3 \\ 0 & 3 & -3 \end{pmatrix} \xrightarrow[R_3 \leftrightarrow R_3 + R_2]{R_2 \leftarrow -\frac{1}{3}R_2} \begin{pmatrix} -2 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{R_1 \leftarrow \frac{1}{2}(R_1 + R_2)} \begin{pmatrix} -1 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

The first equation gives $x = y$, second equation gives $z = y$. Thus, we got eigenvector $v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

$\lambda_{2,3} = 1$

$$(A - I)v = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} v = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} v = 0$$

Thus the eigenvectors are $v_2 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ and $v_3 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$.

d. Write general solution to $x'(t) = Ax(t)$.

Solution: $x(t) = c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^{4t} + c_2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} e^{t} + c_3 \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} e^{t}$

Good luck!